Renal Tumors — I

Pathology GUE Block · MEDucation by Mattin Majid

Continued in Part II: this SGL covers two separate topics — renal tumors (here) and urinary bladder pathology (Part II).

BEFORE YOU READ — ANSWER FROM MEMORY

1. Which benign renal tumor arises from intercalated cells of the collecting duct, and which RCC subtype closely mimics it histologically?
2. What chromosomal abnormality is characteristic of clear cell RCC, and which of phakomatosis/syndrome is it associated with?
3. Name the two genes implicated in Wilms tumor, and the three tissue elements that compose it.

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

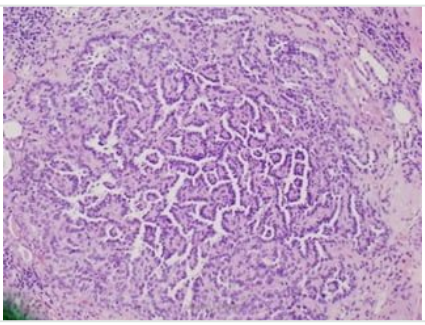
1 Benign (usually) adult tumors

Renal papillary adenoma

Histogenesis	Renal tubular epithelium.
Morphology	Small **<0.5 cm** [>1 cm is indicative of malignancy]; well circumscribed; pale yellow; **cortical** tumors.
Note	**Irrespective of size, all adenomas should be regarded as potentially malignant.**



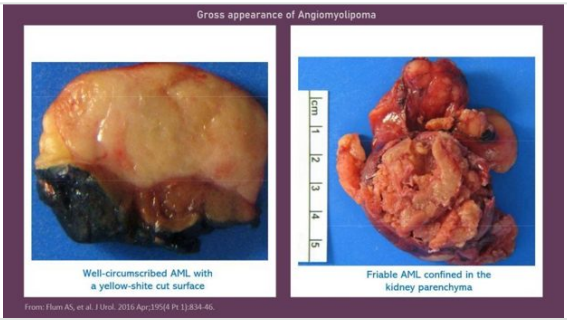
Renal papillary adenoma — gross, cortical



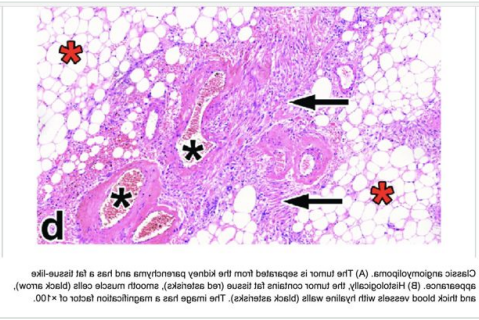
Dense papillary/tubular/tubulopapillary pattern, cuboidal cells

Angiomyolipoma

Nature	**Benign. 25–50% of patients with tuberous sclerosis** (mutation in TSC1 or TSC2 tumor suppressor gene — cerebral cortex lesion, skin abnormalities, epilepsy, intellectual disability, benign tumors elsewhere e.g. heart).
Clinical risk	Increased susceptibility to **spontaneous hemorrhage** because of large size.
Histology	Tumor separated from the kidney parenchyma, has a fat tissue-like appearance; contains fat tissue, **smooth muscle cells**, and thick blood vessels with hyalinized walls.



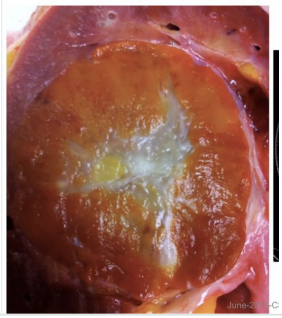
Gross AML — well-circumscribed, yellow-white cut surface; friable AML confined in kidney parenchyma



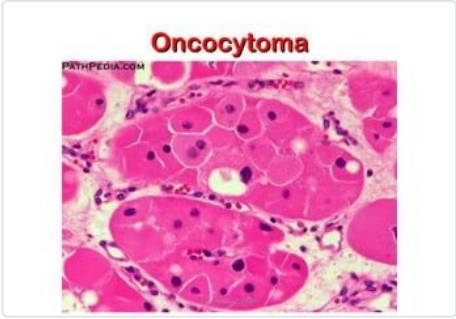
Histology — fat tissue, smooth muscle cells, thick hyalinized blood vessels (x100)

Oncocytoma

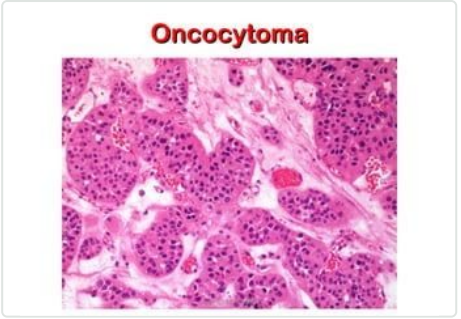
Histogenesis	**Intercalated cells of the collecting duct.**
Gross	**Tan / mahogany brown**; well encapsulated; **central stellate scar**.
Histology	Tumor composed of **large eosinophilic cells**.
EM	**Numerous mitochondria.**



Oncocytoma — mahogany-brown cut surface with central stellate scar



Large eosinophilic cells



Eosinophilic cells with prominent cell borders

2 Malignant tumors

Presenting features: blood in the urine (hematuria), persistent pain in side or back, lump or mass in the side or abdomen, fatigue, unexplained weight loss, loss of appetite, fever.

Paraneoplastic syndrome (RCC): secondary polycythemia (ectopic erythropoietin production). Ectopic production of various hormones or hormone-like substances (ACTH, prolactin, gonadotropins, renin, and parathyroid-like hormone).

Renal cell carcinoma — illness script

WHO	**Adult**; **2:1 male predilection.** Risk factors: smoking, asbestosis, obesity, hypertension, drugs, **ESRD, chronic kidney disease, acquired cystic disease.**
Epidemiology	Most RCC is **sporadic**. Hereditary forms: **VHL syndrome** (sporadic and familial); hereditary leiomyomatosis & RCC (AD); hereditary papillary carcinoma (AD, multiple bilateral tumors with papillary morphology); **Birt-Hogg-Dube syndrome**; a rare variant in young patients is associated with chromosomal translocations involving the **TFE3 gene** at the Xp11.2 locus.

Von Hippel–Lindau disease — associated with: angiomatosis, hemangioblastoma, pheochromocytoma, papillary cystadenoma of epididymis/broad ligament, endolymphatic sac tumor, and RCC.



Von Hippel–Lindau disease — retinal angiomatosis; cerebellar hemangioblastoma on MRI

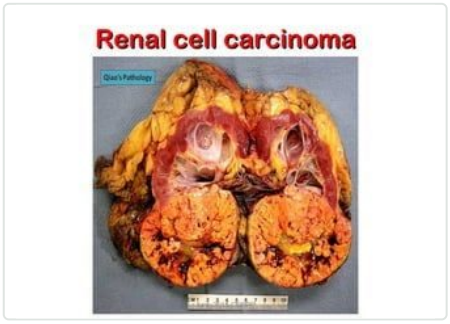


Renal cell carcinoma — gross, bisected kidney

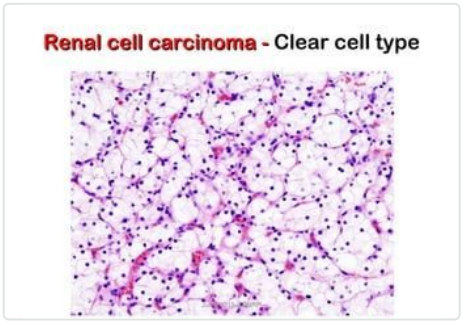
RCC subtypes — the discriminator

Subtype	Key features
(1) Clear cell RCC	**Most common.** Any portion of kidney but commonly poles. **Loss of sequences on the short arm of chromosome 3.** Usually solitary, unilateral. Frequently invades the renal vein or IVC, sometimes as far as the right side of the heart. Gross: bright yellow-grey, spherical mass. Microscopy: round/polygonal cells with abundant clear cytoplasm (glycogen and lipid).
(2) Papillary RCC	Can be **multifocal and bilateral**. **3p deletion.**

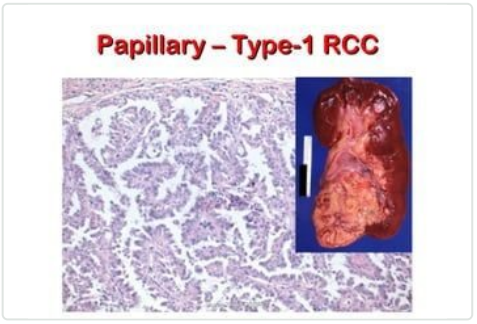
(3) Chromophobe RCC	**Closely mimics oncocytoma**, and like oncocytoma, arises from intercalated cells of collecting ducts. **Excellent prognosis.** Microscopy: eosinophilic granular cells with prominent cell borders and a perinuclear halo, arranged in solid sheets.
(4) Collecting duct (Bellini duct) RCC	**<1%.** Collecting duct cells in the medulla. Microscopy: irregular channels lined by highly atypical epithelium with a **hobnail pattern** within a prominent fibrotic stroma.



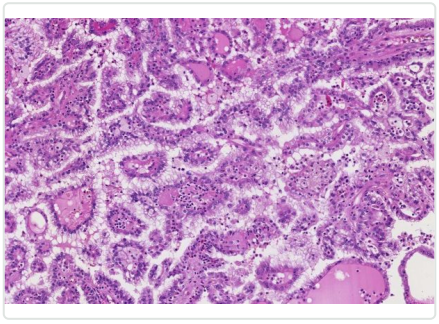
RCC — bright yellow-grey spherical mass



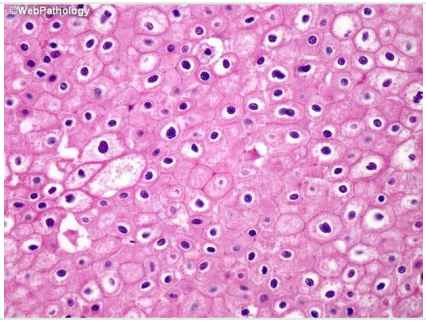
Clear cell RCC — clear cytoplasm (glycogen/lipid)



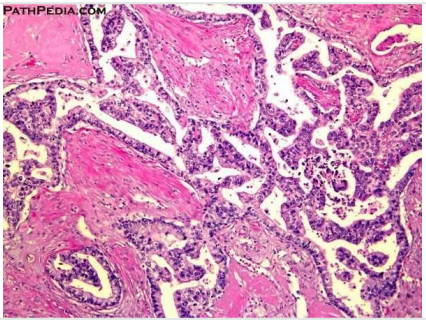
Papillary RCC — papillary architecture



Papillary RCC — gland-forming papillae



Chromophobe RCC — eosinophilic granular cells, perinuclear halo



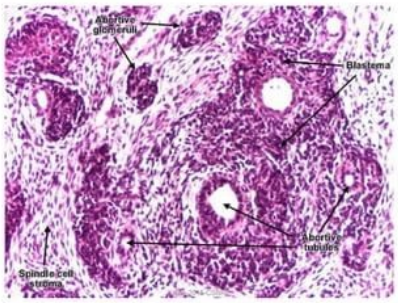
Collecting duct RCC — hobnail epithelium, fibrotic stroma

3 Wilms tumor

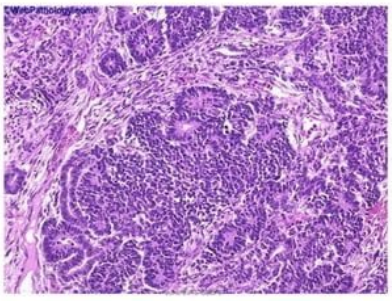
Nature	**Nephroblastoma.** Malignant embryonal tumor; thought to develop from persistent metanephric tissue or nephrogenic rests.
Genetics	**WT1** gene at chromosome **11p13** implicated in tumorigenesis. **WT2** gene at chromosome 11p15 (still not isolated) is a second locus.
Composition	**(Triphasic)** tumor consists of **blastemal, epithelial, and stromal** elements.
Epidemiology	Slight female predominance. Most common: **3–4 years of age.**
What you see	Microscopy: abortive glomeruli and abortive tubules surrounded by blastema and spindle cell stroma.



Wilms tumor — gross, bisected kidney



Abortive glomeruli/tubules



Blastema, spindle cell stroma

P Must-remember pearls

- Irrespective of size, **all renal papillary adenomas are regarded as potentially malignant.**
- Oncocytoma and chromophobe RCC are histologic look-alikes — **both arise from intercalated cells of the collecting duct**; chromophobe has an excellent prognosis despite being malignant.
- Clear cell RCC: **loss of chromosome 3p**, linked to **VHL syndrome**. Papillary RCC: **3p deletion** too, but can be multifocal/bilateral.
- RCC is notorious for **paraneoplastic syndromes** — secondary polycythemia (ectopic EPO) and ectopic ACTH/prolactin/gonadotropin/renin/PTH-like hormone production.
- Clear cell RCC frequently invades the **renal vein or IVC**, occasionally reaching the right atrium.
- Wilms tumor is a **triphasic** tumor (blastemal + epithelial + stromal) peaking at **3–4 years**, driven by **WT1** (11p13) and **WT2** (11p15).
- Angiomyolipoma is linked to **tuberous sclerosis** (TSC1/TSC2) in 25–50% of patients and risks spontaneous hemorrhage from its large size.

Q Test yourself

1. A renal mass shows large eosinophilic cells with numerous mitochondria on EM and a tan/mahogany cut surface with a central stellate scar. What is the diagnosis?

A. Clear cell RCC
B. Oncocytoma
C. Angiomyolipoma
D. Wilms tumor

2. A young woman with tuberous sclerosis is found to have a large renal mass containing fat, smooth muscle, and thick-walled blood vessels. She is at increased risk of:

A. Spontaneous hemorrhage
B. Paraneoplastic polycythemia
C. IVC invasion
D. Distant bone metastasis

3. Renal cell carcinoma with round cells showing abundant clear cytoplasm, loss of chromosome 3p sequences, and frequent renal vein invasion is best classified as:

A. Papillary RCC
B. Chromophobe RCC
C. Clear cell RCC
D. Collecting duct RCC

4. A 3-year-old presents with an abdominal mass. Biopsy shows blastemal, epithelial, and stromal elements with abortive glomeruli and tubules. Which gene is most implicated?

A. VHL
B. WT1
C. TSC1
D. TFE3

5. Which renal tumor subtype is histologically described as closely mimicking oncocytoma, arising from the same cell of origin, with an excellent prognosis?

A. Clear cell RCC
B. Papillary RCC
C. Chromophobe RCC
D. Collecting duct RCC

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 · B — Large eosinophilic cells, numerous mitochondria on EM, tan/mahogany color with central stellate scar describe oncocytoma.

2 · A — Angiomyolipomas carry increased susceptibility to spontaneous hemorrhage because of their large size.

3 · C — Clear cell RCC shows clear cytoplasm (glycogen/lipid), loss of chromosome 3p sequences, and frequently invades the renal vein/IVC.

4 · B — WT1 (chromosome 11p13) is the gene implicated in Wilms tumor (nephroblastoma) tumorigenesis.

5 · C — Chromophobe RCC closely mimics oncocytoma and arises from the same intercalated collecting-duct cells, with an excellent prognosis.

SPACED REVIEW

☐ Day 1	☐ Day 3	☐ Day 7	☐ Day 21	☐ Pre-exam